1956

Poughkeepsie, N. Y.
February 20, 1956

Memorandum to Mr. Lloyd Hubbard

Subject: The Ideal Machine

Enclosed are vague specifications for a wonderful (and probably prohibitively expensive) machine. Following your request for my suggestions I circulated a memo through Programming Research to get other people's opinions also. The enclosed paper is a composite of our ideas of what a machine ought to be.

It is obvious that there is nothing startlingly new here. We are probably too close to the stumps to see the forest. But these comments should give you an idea of what we object to in a machine, even if we have nothing revolutionary to offer.

H. A. Kinslow

HAK:hs

cc: Mr. A. Goble - Poughkeepsie
Mr. W. H. Johnson - WHQ Customer Assistance
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Enc.

THE IDEAL MACHINE WOULD HAVE

Fixed Word Length

With reservations. It should also be possible in some cases at least to operate on several successive words with one order. This need is immediately apparent in input-output and internal transmission. It could be a useful feature in other operations as well.

Command Structure

Operation code - 2 digits
Address - 5 digits

If there are multiple accumulators, or many counters, or several ASU's, this would require the presence of another digit or so to denote them. A command would thus be a minimum of 7 digits, a maximum of perhaps 8 or 9. This would be a large instruction. But we are in general agreement that combinations of zones or digits which have different meanings are pretty horrible. If commands are to have several functions then they should be spread out, and a separate digit allowed for each function. We feel also that commands should be numbers, not letters. But we have no real objection to alphabetic operation codes or alphabetic ASU designations. The address, however, should be numeric, and should be at least 5 digits long.

Half or Full Word Logic

I personally prefer half word logic. If several counters or ASU's are available, their notation would probably stretch the command length to a point where half word logic is unfeasible.

Multiple Accumulators or ASU's

We have had considerable debate as to the value of these things. Many people who have written for the 705 like them, and find them extremely useful. There is universal agreement, however, to the effect that if these things are present in future machines they should by all means be a lot more independent of each other than they are now. The 705 ASU's all share a common sign and zero trigger. This is not exactly an ideal circumstance.

I personally feel that the world can get along nicely without 16 accumulators. They are extremely useful in the 705, but I am inclined to think that a large share of their utility stems from the peculiarities of the 705, and has nothing to do with programming in general.

I am fond of an ACC - MQ arrangement like that on the 701. Many of our people, however, have indicated that they are in favor of extra counters, and provision in the command structure for addressing them. This would probably require full word logic. Is it worth it?

Other General Characteristics

Two people suggested index registers. I was surprised that so little was said about them.

The machine should be decimal,

and single address.

Order Code

The order code should be unambiguous. The operation code should determine the function, not the data. Each operation should have its own discreet function under all circumstances.

Checking

The machine should be self-checking internally, with some attempt to automatically correct errors without programming.

Some people feel that errors should be interrogated by the program. The error signals on the 705, however, have become quite ambiguous. We feel that there should be more error signals to more specifically denote the types of errors that can happen.

Tape reliability needs improvement. We favor double recording, if necessary, with positive checking of readability. The practice of three automatic re-reads or re-writes on errors has become so prevalent we are inclined to think it should be automatic.

Buffers

Each input-output unit should have its own buffer. There should be only one option available, either use the buffer or by-pass it (you can see

that we are becoming satiated with flexibility). Several people suggested that input-output units should be buffered through memory. My only question about this is whether memory can be made large enough. Otherwise I agree.

Specific Machine Functions

Built-in floating point - some people favor this as an optional dial setting. Others would have a programming mode. I decline comment, being relatively unfamiliar with floating point operation.

It should be possible to space a tape without reading it. It should also be possible to read the tape backwards.

A high-speed transmission order, if it exists (which it should) should be limited by information contained in the instruction itself. This, incidentally, is a good rule to apply to any command. As indicated in other paragraphs we are not happy over the fact that the operation of 705 commands depends upon half-a-dozen obscure conditions within the machine. Each command, with the exception of certain input-output operations, should contain within itself all of the information necessary to operate.

We have considerable criticism of the collating sequence. This, I would think, is a peculiarity of the 705.

Pluggable instruction units allowing alternative order codes.

Record counters on each tape unit. These could be used as follows:

a. A TRE would be possible when the tape counter content equals a number N in memory.

b. It would be possible to proceed along the tape, without reading, to a given record N.

Automatic subroutine linkage. This is old hat, but very useful.

Tape compression of data.

The presence somewhere in the system of a control panel for re-arrangement of data as it passes through the machine.

The Console

The console, we feel, is the most neglected piece of equipment. We have many criticisms of present consoles. The console should be the heart of the

whole system. It should contain within it the ability to completely control any unit in the machine without leaving one's chair. We feel strongly about this. The overhead for a machine is the same whether it is operating, or whether you are walking across the floor to space up the paper on the printer. We suggest that the console contain such things as the following:

Addressable dial switches for constants.
An address stop switch (650 type).
Dial switches which could contain an instruction for the machine to execute upon loading.
Decimal information display.
An audible or visible indication that the machine has stopped. If a visible signal, a far more prominent one than those presently used. (Try running the 701 for an eight hour shift sometime, and see how much of a headache you get from watching the console constantly.)

The console should contain all of the buttons, switches and dials which we are at present so inclined to distribute over an acre or so of space. Tape address dials, printer control switches, card punch and card reader controls; all of these should be on the console.

Peripheral operations should be both programmable and controllable from the console. This would mean that a report could be compiled on tape, peripheral printing or punching could begin, and the machine could continue the next phase of an operation simultaneously. Such ability would be extremely valuable in debugging operations. Memory print time - which at present is sheer dead weight - could completely overlap actual test runs.

Notes About Debugging

A great part of a programmer's natural life is spent making very short test runs, trying to squeeze the maximum amount of information out of them, and at the same time working under pressure of time and overhead. There is no more disheartening experience than goofing a test run, and being told, "Sorry, you can't get on the machine again this week". The more complicated a machine setup becomes, the more easily it is goofed. Ergo: it behooves the designer of a machine to dwell a moment upon operational and debugging ease.

Most of our remarks about the console apply here. The 702 and 705 are not easy to operate. They debug grudgingly. Centralization of control would be a great improvement. We would very much like to see steps toward such conveniences as automatic loading, automatic tracing, automatic memory print. Perhaps tracing is a passing fancy. Memory print most certainly is

not. Much of the computer's time is spent spewing its memory. Much of the operator's time is spent copying incorrectly from the console. Would it not be possible to automatically record console information when needed? At present there is no way of even programming this.

Perhaps many of these things are expensive. I suggest that sloppy machine operation is expensive also. I suggest that training people to operate a difficult machine well is more expensive still.